

TS6 - Transformada Z y respuesta en frecuencia

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Dado las siguientes ecuaciones en diferencias de los siguientes sistemas, que representan un filtro de media móvil:

a) $y(n) = x(n-3) + x(n-2) + x(n-1) + x(n)$

b) $y(n) = x(n-4) + x(n-3) + x(n-2) + x(n-1) + x(n)$

c) $y(n) = x(n) - x(n-1)$

d) $y(n) = x(n) - x(n-2)$

Se pide:

1) Hallar $T(z) = \frac{Y(z)}{X(z)}$.

2) Calcular su respuesta en frecuencia de módulo y fase.

3) Simular y validar la respuesta en frecuencia de todos los sistemas con Numpy.

a) $y[n] = x[n-3] + x[n-2] + x[n-1] + x[n]$

$$Y(z) = \sum_{n=-\infty}^{\infty} y[n] z^{-n} = \sum_{n=-\infty}^{\infty} (x[n-3] + x[n-2] + x[n-1] + x[n]) z^{-n}$$

$$\text{como } \sum_{n=-\infty}^{\infty} x[n-n_0] z^{-n} = \sum_{m=-\infty}^{\infty} x[m] z^{-m} \cdot z^{-n_0} = X(z) \cdot z^{-n_0}$$

$m = n - n_0$ $\hookrightarrow X(z)$ por def.

$$\Rightarrow Y(z) = X(z)z^{-3} + X(z)z^{-2} + X(z)z^{-1} + X(z)z^0$$

$$T(z) = \frac{Y(z)}{X(z)} = \frac{X(z)z^{-3} + X(z)z^{-2} + X(z)z^{-1} + X(z)}{X(z)}$$

$$\Rightarrow T(z) = z^{-3} + z^{-2} + z^{-1} + 1 = \frac{z^3}{z^3} (z^{-3} + z^{-2} + z^{-1} + 1) = \frac{1 + z + z^2 + z^3}{z^3}$$

$$\Rightarrow T(z) = \frac{z^3 + z^2 + z + 1}{z^3}$$

$\hookrightarrow$ Polo en el origen. FIR

$$z = re^{j\omega} \text{ si } r=1 \Rightarrow T(z) \Big|_{z=e^{j\omega}} = \frac{(e^{j\omega})^3 + (e^{j\omega})^2 + e^{j\omega} + 1}{(e^{j\omega})^3} = e^{-j3\frac{\omega}{2}} \frac{(e^{j\frac{\omega}{2}} + e^{-j\frac{\omega}{2}})}{e^{j\frac{\omega}{2}}} + e^{-j\frac{\omega}{2}} \frac{(e^{j\frac{\omega}{2}} + e^{-j\frac{\omega}{2}})}{e^{j\frac{\omega}{2}}} =$$

$$= e^{-j\frac{\omega}{2}} (e^{j\frac{\omega}{2}} + e^{-j\frac{\omega}{2}}) + e^{-j\frac{3\omega}{2}} (e^{j\frac{\omega}{2}} + e^{-j\frac{\omega}{2}})$$

$$= (e^{-j\frac{\omega}{2}} + e^{-j\frac{3\omega}{2}}) (e^{j\frac{\omega}{2}} + e^{-j\frac{\omega}{2}})$$

$$= e^{-j\frac{3\omega}{2}} (e^{j\omega} + e^{-j\omega}) (e^{j\frac{\omega}{2}} + e^{-j\frac{\omega}{2}})$$

$$\cos(\alpha) = \frac{e^{j\alpha} + e^{-j\alpha}}{2}$$

$$= e^{-j\frac{3\omega}{2}} \cdot 2 \cdot \cos(\omega) \cdot 2 \cdot \cos(\omega/2)$$

$$= 4 \cos(\omega) \cos(\omega/2) e^{-j\frac{3\omega}{2}}$$

$$\cos(\alpha) \cos(\beta) = \frac{1}{2} [\cos(\alpha + \beta) + \cos(\alpha - \beta)]$$

$$= 4 \cdot \frac{1}{2} [\cos(3\omega/2) + \cos(\omega/2)] e^{-j\frac{3\omega}{2}}$$

$$T(\omega) = 2 \cdot [\cos(3\omega/2) + \cos(\omega/2)] e^{-j\frac{3\omega}{2}}$$

$\hookrightarrow T_R$ $\hookrightarrow 4 T(\omega)$

$$|T(\omega)| = |T_R|$$

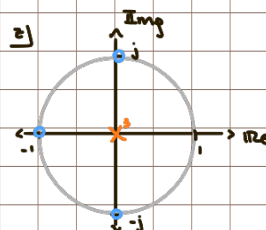
$$\Rightarrow T(z) = \frac{z^3 + z^2 + z + 1}{z^3} \rightarrow 3 \text{ ceros: } z = -1, z = j, z = -j$$

$\hookrightarrow 3$ polos en el origen

$$z^3 + z^2 + z + 1 = 0 \quad z^2 + 1 = 0 \quad z + 1 = 0$$

$$z^2(z+1) + 1(z+1) \quad z^2 = -1 \quad z = -1$$

$$(z^2+1)(z+1) = 0 \quad z = \pm j$$

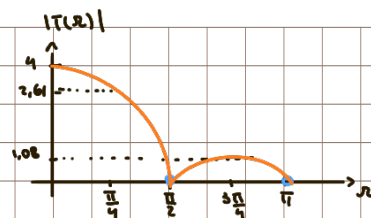


no Diagrama de polos y ceros

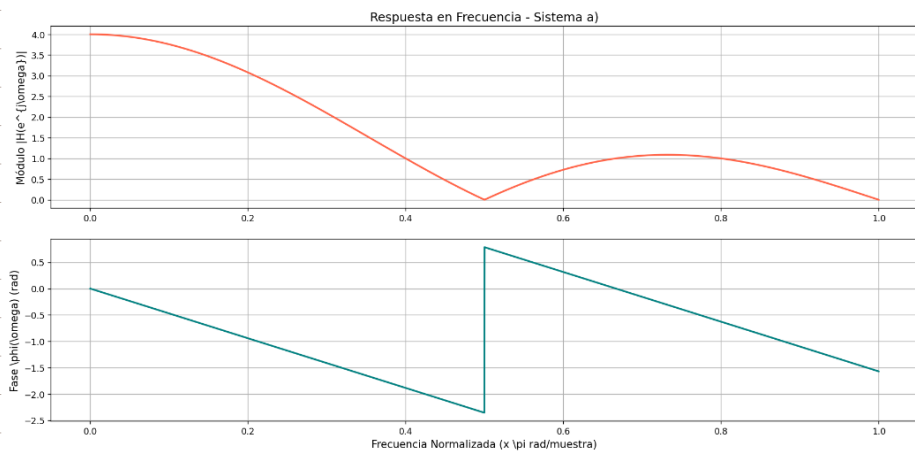
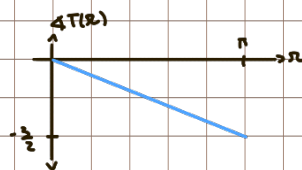
$$|T(z)|_{z=0} = 2 \cdot [\cos(0) + \cos(0)] = 4$$

$$|T(z)|_{z=\pi/4} = 2 \cdot [\cos(3\pi/8) + \cos(\pi/8)] \approx 2,61$$

$$|T(z)|_{z=3\pi/4} = 2 \cdot [\cos(9\pi/8) + \cos(3\pi/8)] \approx 1,08$$



$4T(z) = -\frac{z}{2}$ no tiene de pendiente $-1/2$ y ordenada en el origen



b) $y[n] = x[n-4] + x[n-3] + x[n-2] + x[n-1] + x[n]$

$$\Rightarrow Y(z) = X(z)z^{-4} + X(z)z^{-3} + X(z)z^{-2} + X(z)z^{-1} + X(z)$$

$$\Rightarrow T(z) = \frac{Y(z)}{X(z)} = \frac{X(z)z^{-4} + X(z)z^{-3} + X(z)z^{-2} + X(z)z^{-1} + X(z)}{X(z)}$$

$$T(z) = z^{-4} + z^{-3} + z^{-2} + z^{-1} + 1 = \frac{1 + z + z^2 + z^3 + z^4}{z^4}$$

$$\Rightarrow T(z) = \frac{z^4 + z^3 + z^2 + z + 1}{z^4} \rightarrow 4 \text{ polos en el origen}$$

$$\begin{aligned} T(z)_{z=e^{j2\pi}} &= \frac{e^{j4\pi} + e^{j3\pi} + e^{j2\pi} + e^{j\pi} + 1}{e^{j4\pi}} \\ &= \frac{e^{j2\pi}}{e^{j4\pi}} (e^{j2\pi} + e^{j\pi} + 1 + e^{-j\pi} + e^{-j2\pi}) \\ &= e^{-j2\pi} (e^{j2\pi} + e^{j\pi} + 1 + e^{-j2\pi} + e^{-j\pi}) \\ &= e^{-j2\pi} (2 \cos(2\pi) + 1 + 2 \cos(2\pi)) \end{aligned}$$

$$\Rightarrow T(z) = \underbrace{[1 + 2 \cos(z) + 2 \cos(2z)]}_{T_0} \underbrace{e^{j2\pi}}_{4T(z)}$$

$$|T(z)| = |T_0| = |1 + 2 \cos(z) + 2 \cos(2z)|$$

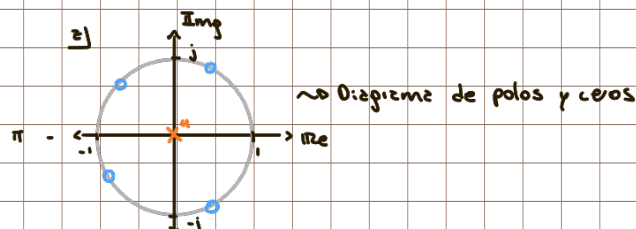
$$\Rightarrow |T(z)|_{z=0} = 1 + 4 = 5$$

$$|T(z)|_{z=\pi/2} = |1 + 2 \cdot 0 + 2 \cdot (-1)| = 1$$

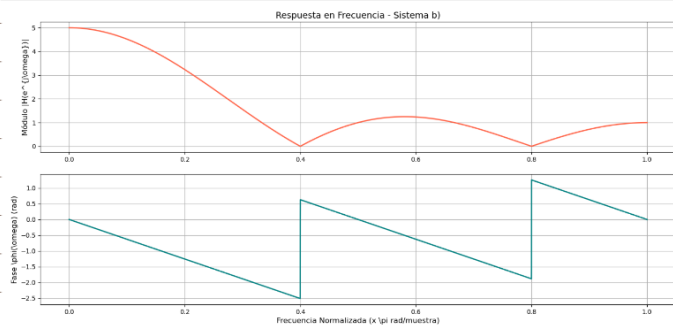
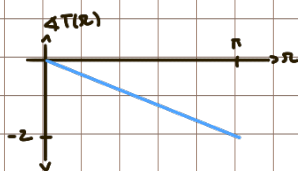
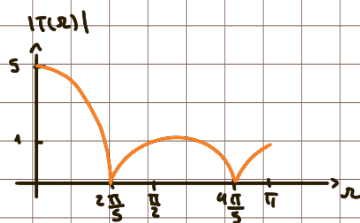
$$|T(z)|_{z=\pi} = |1 + 2 \cdot (-1) + 2 \cdot 1| = 1$$

$$\begin{aligned} z^4 + z^3 + z^2 + z + 1 &= 0 \\ (z-1)(z^4 + z^3 + z^2 + z + 1) &= (z-1) \cdot 0 \\ (z^5 + z^4 + z^3 + z^2 + z) - (z^4 + z^3 + z^2 + z + 1) &= 0 \\ z^5 - 1 &= 0 \\ z^5 &= 1 / \text{como } 1 = e^{j2\pi k} \\ z &= e^{j\frac{2\pi k}{5}} \end{aligned}$$

$$\begin{aligned} \Rightarrow k=0 &\rightarrow z_0 = e^0 = 1 \rightarrow \text{la del } (z-1) = 0, \text{ la descarto} \\ k=1 &\rightarrow z_1 = e^{j\frac{2\pi}{5}} \\ k=2 &\rightarrow z_2 = e^{j\frac{4\pi}{5}} \\ k=3 &\rightarrow z_3 = e^{j\frac{6\pi}{5}} \\ k=4 &\rightarrow z_4 = e^{j\frac{8\pi}{5}} \end{aligned}$$



no Diagrama de polos y ceros



c) $y[n] = x[n] - x[n-1]$

$$Y(z) = X(z) - X(z)z^{-1}$$

$$T(z) = \frac{Y(z)}{X(z)} = \frac{X(z) - X(z)z^{-1}}{X(z)} = 1 - z^{-1} = \frac{z-1}{z}$$

$\Rightarrow T(z) = \frac{z-1}{z}$ → 1 cero en $z=1$
→ 1 polo en el origen

$$T(z) \Big|_{z=e^{j\omega}} = \frac{e^{j\omega} - 1}{e^{j\omega}} = e^{j\frac{\omega}{2}} \frac{(e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}})}{e^{j\frac{\omega}{2}}} = e^{j\frac{\omega}{2}} \cdot 2j \sin(\omega/2) = e^{-j\frac{\omega}{2}} \cdot e^{j\frac{\omega}{2}} \cdot 2 \sin(\omega/2)$$

como $j = e^{j\frac{\pi}{2}}$

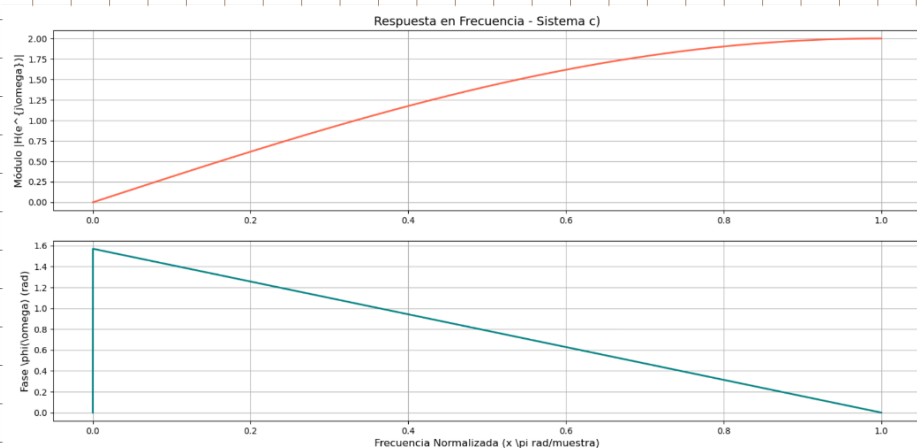
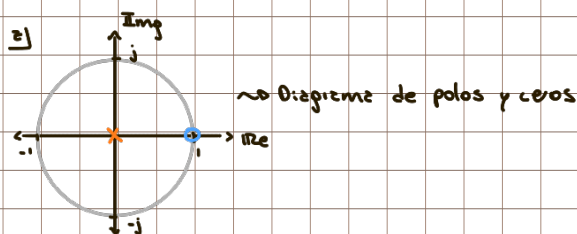
$\Rightarrow T(\omega) = 2 \sin(\omega/2) \cdot e^{j0}$
→ T_R → $\phi_T(\omega)$

como $|T(\omega)| = |T_R|$

$\Rightarrow |T(\omega)| \Big|_{\omega=0} = |2 \cdot 0| = 0 \rightarrow$ por el cero en $z=1$

$$|T(\omega)| \Big|_{\omega=\frac{\pi}{2}} = \left| 2 \cdot \frac{\sqrt{2}}{2} \right| = \sqrt{2}$$

$$|T(\omega)| \Big|_{\omega=\pi} = |2 \cdot 1| = 2$$



d) $y[n] = x[n] - x[n-2]$

$$Y(z) = X(z) - X(z)z^{-2}$$

$$T(z) = \frac{Y(z)}{X(z)} = \frac{X(z) - X(z)z^{-2}}{X(z)} = 1 - z^{-2} = \frac{z^2 - 1}{z^2}$$

$\Rightarrow T(z) = \frac{z^2 - 1}{z^2}$
→ 2 ceros $z^2 - 1 = 0$ $z^2 = 1 \Rightarrow z = -1, z = 1$
→ 2 polos en el origen

$$T(z) \Big|_{z=e^{j\omega}} = \frac{e^{j2\omega} - 1}{e^{j2\omega}} = e^{j\omega} \frac{(e^{j\omega} - e^{-j\omega})}{e^{j2\omega}} = e^{-j\omega} 2j \sin(\omega) = e^{-j\omega} e^{j\omega/2} 2 \sin(\omega/2) = e^{-j\omega/2} \cdot 2 \sin(\omega/2)$$

$\Rightarrow T(\omega) = \underbrace{2 \sin(\omega/2)}_{|T_p|} \underbrace{e^{-j\omega/2}}_{\angle T(\omega)}$

como $|T(\omega)| = |T_p|$

$$|T(\omega)| \Big|_{\omega=0} = |2 \sin(0)| = 0$$

$$|T(\omega)| \Big|_{\omega=\pi/2} = |2 \sin(\pi/4)| = 2$$

$$|T(\omega)| \Big|_{\omega=\pi} = |2 \sin(\pi/2)| = 0$$

